```
while (true)
{
  if(rax ≤ 5) break;
  rax = (rax ∧ 3) + rax >> 2
}
rax = rax % 3;
```

$/2^2$ [0x23]

$\%4$

is mod 3

```
while (true)
{
  if (rax ≤ 5) break
  rax = ((rax ∧ 3) + rax >> 2)
}
```

rax / 2^2

rdx

is $rax \in \{3,4,5\} \Rightarrow CF = 0$ → cmc → $CF = 1$

$CF = 1 \Leftarrow rax < 3$

$CF = 0$

$rdx = rdx - ((rdx + CF) \wedge 3)$ → 0

$rdx = rdx - ((0) + 1 \wedge 3)$

is $rdx = rdx - 1$

$rax = rax - rdx$ (5, 3)

$rdx \geq 3 \Rightarrow rdx \in \{0,1,2,3\}$

mod 5?

```
.loop:
        cmp     rax, 5
        jbe     .exit_loop
        mov     rdx, rax
        shr     rdx, 2
        and     rax, 3
        add     rax, rdx
        jmp     .loop
.exit_loop:
        cmp     rax, 3
        cmc     "complement carry flag"
        sbb     rdx, rdx
        and     rdx, 3
        sub     rax, rdx
```

$rax \leq 5 \Rightarrow$ exit loop

ELSE

$rdx := rax > 5$

$rdx = rax / 2^2$

$rax = rax \wedge 0011$

$rax = rax + rdx$

is loop:

$rax = (rax \wedge 0011) + rax / 2^2$

$rdx = rdx - (rdx + CF)$

$rdx = rdx \wedge 3$

$rax = rax - rdx$

$= rdx - (rdx \wedge 3)$

$= rdx - ((rdx - rdx + CF) \wedge 3)$

$ZF := 1 \Leftrightarrow rax = 3$

$SF := 1 \Leftrightarrow rax < 3$

$CF := 1$

mod 3

is $\neg CF$

```
.loop:
    cmp     rax, 5
    jbe     .exit_loop
    mov     rax, rax
    shr     rdx, 2
    and     rdx, 3
    add     rax, rdx
    jmp     .loop
```

(rax − 5) accumulator

> 5: $SF, CF == 0$

$= 5$: $ZF = 1$

< 5: $CF = 1$

"jump if below or equal" $\Leftrightarrow$ CF or $ZF = 1$

rax → rdx is $rdx := rax$ (accumulator)

rdx is rdx / 2^2

rdx = {0, 1, 2, 3}

rax = rax + rdx is rax = {0, 1, 2, 3}

unconditional

```
.exit_loop:
    cmp     rax, 3
    cmc     "complement carry flag"
    sbb     rdx, rdx
    and     rdx, 3
    sub     rax, rdx
```

is $CF = 1 \rightarrow CF = 0$

$CF = 0 \rightarrow CF = 1$

rdx $\xrightarrow{CF=0}$ 0 $\xrightarrow{CF=1}$ 0xF···F ~ 1111 ~ −1

rdx := {0, 1, 2, 3}

rax := rax − rdx

= rax − {0, 1, 2, 3}

?